Database Tables Specification

GIU Food Truck
Management System

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Overview: This document provides a comprehensive list of all database tables in the GIU Food Truck Management System, including their columns, primary keys, foreign keys, and relationship attributes.

1 Entity-Relationship Diagram

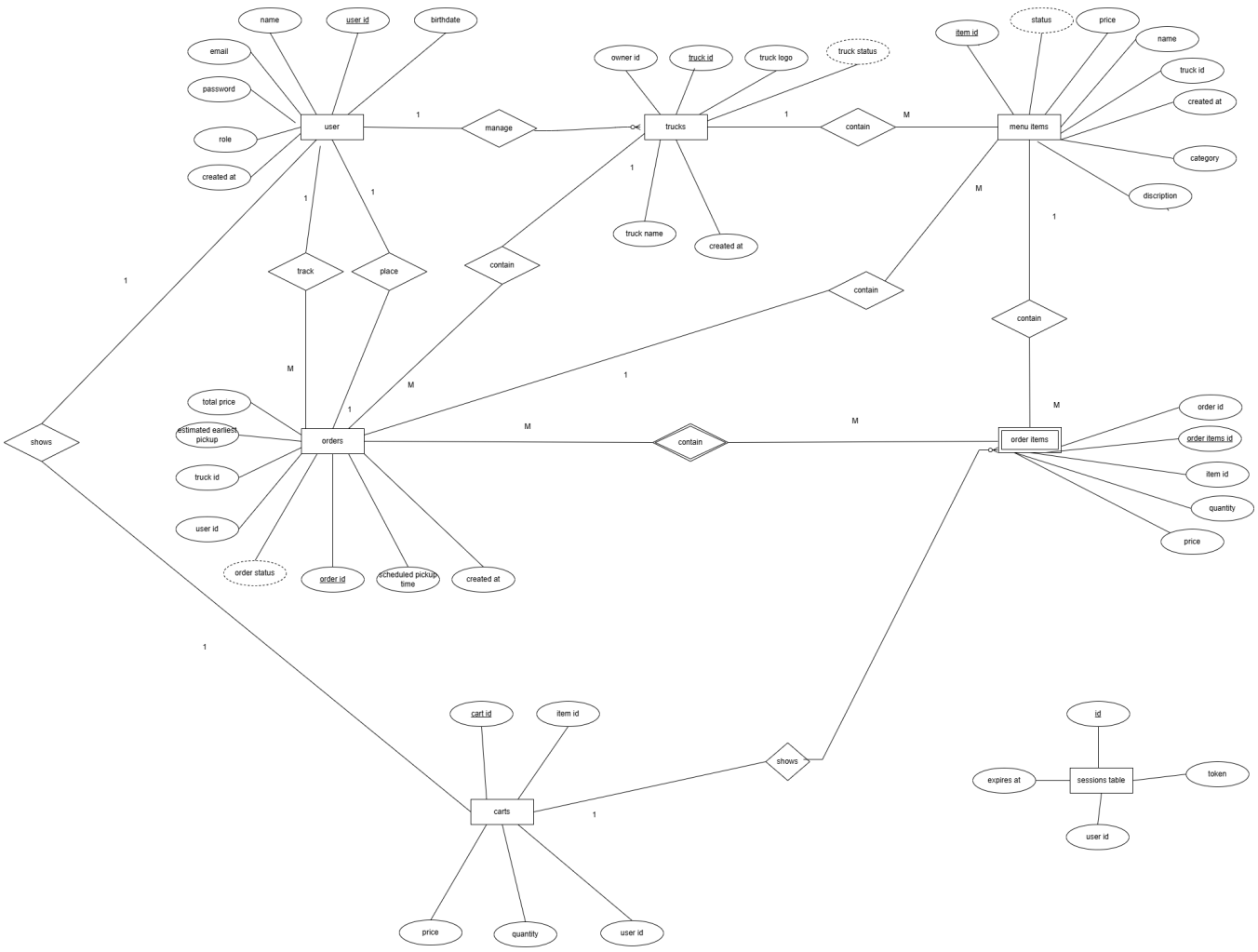

Figure 1: GIU Food Truck Management System - Entity-Relationship Diagram

2 Database Tables Overview

This document describes the complete database schema with **7 Entity Tables** and **8 Junction Tables** (15 tables total).

2.1 Entity Tables

The following 7 tables represent the core entities in the system:

2.1.1 1. Users Table

Table Name: `FoodTruck.Users`
Description: Stores information about system users (customers and truck owners).

Column Name	Data Type	Description
userId	serial	**Primary Key** - Unique identifier for each user
name	text	User's full name (NOT NULL)
email	text	User's email address (NOT NULL, UNIQUE)
password	text	Hashed password (NOT NULL)
role	text	User role: 'customer' or 'truckOwner' (DEFAULT: 'customer')
birthDate	date	User's date of birth (DEFAULT: current_timestamp)
createdAt	timestamp	Account creation timestamp (DEFAULT: current_timestamp)

2.1.2 2. Trucks Table

Table Name: `FoodTruck.Trucks`
Description: Stores information about food trucks.

Column Name	Data Type	Description
truckId	serial	**Primary Key** - Unique identifier for each truck
truckName	text	Name of the food truck (NOT NULL, UNIQUE)
truckLogo	text	URL to truck logo image
ownerId	integer	**Foreign Key** references Users(userId) - Owner of the truck (NOT NULL)
truckStatus	text	Operating status: 'available', 'unavailable' (DEFAULT: 'available')
orderStatus	text	Order acceptance status: 'available', 'unavailable' (DEFAULT: 'available')
createdAt	timestamp	Truck registration timestamp (DEFAULT: current_timestamp)

Foreign Key Constraints:

- ownerId REFERENCES Users(userId) ON DELETE CASCADE

2.1.3 3. MenuItems Table

Table Name: `FoodTruck.MenuItems`
Description: Stores menu items for each food truck.

Column Name	Data Type	Description
itemId	serial	**Primary Key** - Unique identifier for each menu item
truckId	integer	**Foreign Key** references Trucks(truckId) - Associated truck (NOT NULL)
name	text	Name of the menu item (NOT NULL)
description	text	Detailed description of the item
price	numeric(10,2)	Price of the item (NOT NULL)
category	text	Category: 'Main Course', 'Sides', 'Beverages', etc. (NOT NULL)
status	text	Availability status: 'available', 'unavailable' (DEFAULT: 'available')
createdAt	timestamp	Item creation timestamp (DEFAULT: current_timestamp)

Foreign Key Constraints:

- truckId REFERENCES Trucks(truckId) ON DELETE CASCADE

2.1.4 4. Orders Table

Table Name: `FoodTruck.Orders`
Description: Stores customer orders.

Column Name	Data Type	Description
orderId	serial	**Primary Key** - Unique identifier for each order

Column Name	Data Type	Description
userId	integer	**Foreign Key** references Users(userId) - Customer who placed the order (NOT NULL)
truckId	integer	**Foreign Key** references Trucks(truckId) - Truck fulfilling the order (NOT NULL)
orderStatus	text	Order status: 'pending', 'confirmed', 'ready', 'completed', 'cancelled' (NOT NULL)
totalPrice	numeric(10,2)	Total price of the order (NOT NULL)
scheduledPickup-Time	timestamp	Customer's preferred pickup time
estimated-EarliestPickup	timestamp	Estimated earliest pickup time
createdAt	timestamp	Order creation timestamp (DEFAULT: current_timestamp)

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- truckId REFERENCES Trucks(truckId) ON DELETE CASCADE

2.1.5 5. OrderItems Table

Table Name: `FoodTruck.OrderItems`
Description: Stores individual line items within an order. Each record represents one menu item with its quantity and price in an order.

Column Name	Data Type	Description
orderItemId	serial	**Primary Key** - Unique identifier for each order item
name	text	Name/description of the ordered item (NOT NULL)
quantity	integer	Quantity ordered (NOT NULL)
price	numeric(10,2)	Unit price at time of order (NOT NULL)
createdAt	timestamp	When the item was added (DEFAULT: current_timestamp)

2.1.6 6. Carts Table

Table Name: `FoodTruck.Carts`
Description: Represents items in a user's shopping cart. Each record links a user to a menu item with quantity and price.

Column Name	Data Type	Description
cartId	serial	**Primary Key** - Unique identifier for each cart item
userId	integer	**Foreign Key** references Users(userId) - User who owns this cart item (NOT NULL)
itemId	integer	**Foreign Key** references MenuItems(itemId) - Menu item in cart (NOT NULL)
quantity	integer	Quantity of this item in cart (NOT NULL)
price	numeric(10,2)	Price of the item (NOT NULL)

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- itemId REFERENCES MenuItems(itemId) ON DELETE CASCADE

2.1.7 7. Sessions Table

Table Name: `FoodTruck.Sessions`
Description: Stores user authentication sessions.

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier for each session
userId	integer	**Foreign Key** references Users(userId) - User associated with session (NOT NULL)
token	text	Authentication token (NOT NULL)
expiresAt	timestamp	Session expiration timestamp (DEFAULT: current_timestamp)

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE

2.2 Junction Tables

The following 8 junction tables implement the many-to-many and one-to-many relationships between entities:

2.2.1 1. User_View_Cart Table

Table Name: `FoodTruck.User_View_Cart`
Description: Junction table implementing the 1:1 relationship between User and Cart. Each user views exactly one cart. **Cardinality:** 1:1 (User ↔ Cart)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
userId	integer	**Foreign Key** references Users(userId) - User viewing cart (NOT NULL, UNIQUE)
cartId	integer	**Foreign Key** references Carts(cartId) - Cart being viewed (NOT NULL, UNIQUE)
createdAt	timestamp	When relationship was established (DEFAULT: current_timestamp)

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- cartId REFERENCES Carts(cartId) ON DELETE CASCADE
- UNIQUE(userId) - ensures one cart per user
- UNIQUE(cartId) - ensures one user per cart

2.2.2 2. User_Track_Order Table

Table Name: `FoodTruck.User_Track_Order`
Description: Junction table implementing the 1:M relationship between User and Orders. Tracks which orders belong to which user.
Cardinality: 1:M (User → Orders)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
userId	integer	**Foreign Key** references Users(userId) - User tracking order (NOT NULL)
orderId	integer	**Foreign Key** references Orders(orderId) - Order being tracked (NOT NULL)
lastViewed	timestamp	Last time user viewed order status (DEFAULT: current_timestamp)
notificationsEnable	boolean	Whether user wants order notifications (DEFAULT: true)

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- orderId REFERENCES Orders(orderId) ON DELETE CASCADE

2.2.3 3. User_Place_Order Table

Table Name: `FoodTruck.User_Place_Order`
Description: Junction table implementing the 1:1 relationship for the action of placing an order. Captures the moment a user submits an order.
Cardinality: 1:1 (User → Order placement action)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
userId	integer	**Foreign Key** references Users(userId) - User placing order (NOT NULL)
orderId	integer	**Foreign Key** references Orders(orderId) - Order being placed (NOT NULL, UNIQUE)
placedAt	timestamp	Exact timestamp when order was placed (NOT NULL)
ipAddress	text	IP address from which order was placed
deviceInfo	text	Device information for order placement

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- orderId REFERENCES Orders(orderId) ON DELETE CASCADE
- UNIQUE(orderId) - each order placed exactly once

2.2.4 4. User_Manage_Trucks Table

Table Name: `FoodTruck.User_Manage_Trucks`
Description: Junction table implementing the 1:M relationship between truck owners and their trucks. Links users with role='truckOwner' to the trucks they manage.
Cardinality: 1:M (User → Trucks)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
userId	integer	**Foreign Key** references Users(userId) - Truck owner (NOT NULL)
truckId	integer	**Foreign Key** references Trucks(truckId) - Managed truck (NOT NULL)
assignedAt	timestamp	When ownership/management began (DEFAULT: current_timestamp)
permissions	text	Management permissions level (DEFAULT: 'full')

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- truckId REFERENCES Trucks(truckId) ON DELETE CASCADE

2.2.5 5. User_AddRemove_MenuItems Table

Table Name: `FoodTruck.User_AddRemove_MenuItems`
Description: Junction table implementing the 1:M relationship between truck owners and menu items. Tracks which user added/modified which menu items.
Cardinality: 1:M (User → MenuItems)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
userId	integer	**Foreign Key** references Users(userId) - User performing action (NOT NULL)
itemId	integer	**Foreign Key** references MenuItems(itemId) - Menu item affected (NOT NULL)
action	text	Action performed: 'add', 'update', 'remove' (NOT NULL)
actionTimestamp	timestamp	When action was performed (DEFAULT: current_timestamp)
notes	text	Optional notes about the action

Foreign Key Constraints:

- userId REFERENCES Users(userId) ON DELETE CASCADE
- itemId REFERENCES MenuItems(itemId) ON DELETE CASCADE

2.2.6 6. Truck_Contains_MenuItems Table

Table Name: `FoodTruck.Truck_Contains_MenuItems`
Description: Junction table implementing the 1:M relationship between trucks and their menu items. Links each truck to its menu.
Cardinality: 1:M (Truck → MenuItems)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
truckId	integer	**Foreign Key** references Trucks(truckId) - Truck containing item (NOT NULL)
itemId	integer	**Foreign Key** references MenuItems(itemId) - Menu item in truck (NOT NULL)
displayOrder	integer	Order in which item appears on menu (DEFAULT: 0)
addedAt	timestamp	When item was added to truck menu (DEFAULT: current_timestamp)
isFeatured	boolean	Whether item is featured on menu (DEFAULT: false)

Foreign Key Constraints:

- truckId REFERENCES Trucks(truckId) ON DELETE CASCADE
- itemId REFERENCES MenuItems(itemId) ON DELETE CASCADE

2.2.7 7. Order_Contains_MenuItems Table

Table Name: `FoodTruck.Order_Contains_MenuItems`
Description: Junction table implementing the M:N relationship between orders and menu items. Links orders to the specific menu items ordered.
Cardinality: M:N (Orders ↔ MenuItems)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
orderId	integer	**Foreign Key** references Orders(orderId) - Order containing items (NOT NULL)
itemId	integer	**Foreign Key** references MenuItems(itemId) - Menu item in order (NOT NULL)
quantity	integer	Quantity of this menu item ordered (NOT NULL)

Column Name	Data Type	Description
priceAtOrder	numeric(10,2)	Price of menu item at time of order (NOT NULL)

Foreign Key Constraints:

- orderId REFERENCES Orders(orderId) ON DELETE CASCADE
- itemId REFERENCES MenuItems(itemId) ON DELETE CASCADE

2.2.8 8. Order_Contains_OrderItems Table

Table Name: `FoodTruck.Order_Contains_OrderItems`
Description: Junction table implementing the 1:M relationship between orders and order items. Links each order to its constituent line items.
Cardinality: 1:M (Order → OrderItems)

Column Name	Data Type	Description
id	serial	**Primary Key** - Unique identifier
orderId	integer	**Foreign Key** references Orders(orderId) - Parent order (NOT NULL)
orderItemId	integer	**Foreign Key** references OrderItems(orderItemId) - Order item (NOT NULL)
lineNumber	integer	Line number in the order (for display purposes)
createdAt	timestamp	When item was added to order (DEFAULT: current_timestamp)

Foreign Key Constraints:

- orderId REFERENCES Orders(orderId) ON DELETE CASCADE
- orderItemId REFERENCES OrderItems(orderItemId) ON DELETE CASCADE

3 Entity-Relationship Summary

3.1 Entity Tables (7 Total)

1. **Users** - User accounts (customers and truck owners)
2. **Trucks** - Food truck information
3. **MenuItems** - Menu items available for purchase
4. **Orders** - Customer orders
5. **OrderItems** - Individual line items within orders
6. **Carts** - Shopping cart entities
7. **Sessions** - Authentication sessions

3.2 Junction Tables (8 Total)

1. **User_View_Cart** - User ↔ Cart (1:1)
2. **User_Track_Order** - User → Orders (1:M)
3. **User_Place_Order** - User → Order (1:1)
4. **User_Manage_Trucks** - User → Trucks (1:M)
5. **User_AddRemove_MenuItems** - User → MenuItems (1:M)
6. **Truck_Contains_MenuItems** - Truck → MenuItems (1:M)
7. **Order_Contains_MenuItems** - Order ↔ MenuItems (M:N)
8. **Order_Contains_OrderItems** - Order → OrderItems (1:M)

3.3 Complete Relationship Summary

User Relationships:

- **User ↔ Cart (1:1):** Each user views one cart
- **User → Orders (1:M):** User tracks multiple orders
- **User → Order (1:1):** User places order (action)
- **User → Trucks (1:M):** Truck owner manages multiple trucks
- **User → MenuItems (1:M):** Owner adds/removes menu items

Truck Relationships:

- **Truck → MenuItems (1:M):** Truck contains multiple menu items
- **Truck → Orders (1:M):** Truck fulfills multiple orders

Order Relationships:

- **Order ↔ MenuItems (M:N):** Orders contain menu items
- **Order → OrderItems (1:M):** Order contains multiple line items

Total Tables: 7 Entity Tables + 8 Junction Tables = **15 Tables**

4 Database Schema Notes

- All tables use `serial` for primary keys (auto-incrementing integers)
- Foreign key constraints include `ON DELETE CASCADE` for referential integrity
- Timestamps use PostgreSQL `timestamp` data type with `DEFAULT current_timestamp`
- Price fields use `numeric(10,2)` for precise decimal handling
- All user-related data is deleted when a user account is removed (cascade delete)
- Email addresses are unique across the system
- Truck names are unique across the system

End of Database Tables Specification

GIU Food Truck Management System